Core distributed semantics: specification and formal verification

Created-at: 2026-09-05 17:10 CEST

Scope. Continue the user's distributed execution investigation, update its existing single-writer/replica semantics, and verify concrete safety properties. This does not imply multi-writer consensus, an implemented scheduler, or unrestricted algebraic reassociation. The earlier dependency/CI, browser and consumer stages remain in scope for subsequent integration; Amaterasu is authorized.

ETA. 45–90 minutes: public-API counterexamples, a small Lean model using the already installed 4.19.0 toolchain, specification corrections, traceability, review and report. Individual verification commands are bounded and logged.

Source and ownership. Own Core worktree D:/hnnwork/distributed/core, branch audit/core-distributed-semantics-2026-09-05, base 4a243ede85b7dd1f488bc9c68a46086a8638e39c. Core owns the IR, portable codec, version tuple and transaction assembler. Chordex owns opaque transport. Consumers own computation scheduling and installation, with domain algorithms in their existing framework owners. The existing Core specification and docs/verification/unified-ir remain the canonical location. The historical proofs directory is empty; unrelated SymPy/Z3 algebra proofs are not evidence of distributed replay correctness.

Discovery. Read unified-ir-architecture Part G, hypergraph-synchronization, requirements, SnapshotVersion, TransactionAssembler and existing test scaffolding. Reuse the public EnvelopeDraft/encode/validate/apply boundary. Do not duplicate assembler or IR code in a worker. Lean will model only the commit admission boundary, with explicitly stated abstraction and refinement limits.

Files. Modify docs/unified-ir-architecture.md, docs/hypergraph-synchronization.md and docs/verification/unified-ir/requirements.md. New: docs/verification/unified-ir/distributed-semantics.lean; docs/plans/2026-09-05-distributed-semantics/plan.tex, plan.pdf, plan.tikz, plan.mmd; reports/2026-09-05-distributed-semantics.md and reports/distributed-semantics/ (Rust probe, logs, traceability and hashes). Extend .gitattributes only for exact evidence bytes. No new Cargo dependency.

Protected changes. Initial protected Core source edits: empty list. First reproduce admission gaps through the public facade and prepare a concrete minimal correction with regression/rollback details. The Core manifest requires a scoped token for a protected implementation edit; existing Graph approvals do not name this code. Independent specification and proof work proceeds without such edits.

Verification. Check version-order laws, non-regression, equal-version unknown-identity rejection, stale delta rejection, replay identity, and conditional replica convergence. Kernel-check Lean theorems without sorry or added axioms; record their actual axiom dependencies. A proof about the abstract model is not a proof of Rust refinement, payload validation, durable crash recovery or liveness. Run an independent Rust probe against the exact containerized Core on Amaterasu; include negative controls that expose implementation violations rather than changing expected outcomes to obtain a green acceptance result.

Budgets and risks. At most 16 GiB RSS, no swap/virtual-memory cap. Remote probes use the existing bounded Docker SDK, network disabled. Lean uses the existing 2 GiB sampled RSS guard; no package download. The installed 4.19.0 rejects the attempted internal memory option, so that failed attempt is retained rather than counted as a cap. Capture exit, wall time and RSS; sample-based RSS limits are not a kernel quota. No throughput claim from tiny probes. Risks include missing persisted identities, incomparable version tuples, payload-vs-structural-hash confusion, and unsupported claims of exactly-once work or atomic installation from transport receipt alone.

Acceptance and rollback. Every requirement separates normative model proof from observed runtime status. Preserve and report any red runtime acceptance check. Keep the prior audit and Docker evidence unchanged. Local thematic commits are authorized by the continuing task; no push, merge, release or global host changes. Roll back only this branch's named changes, retaining failure evidence needed for review.

2026-09-13: concurrent preparation and local publication

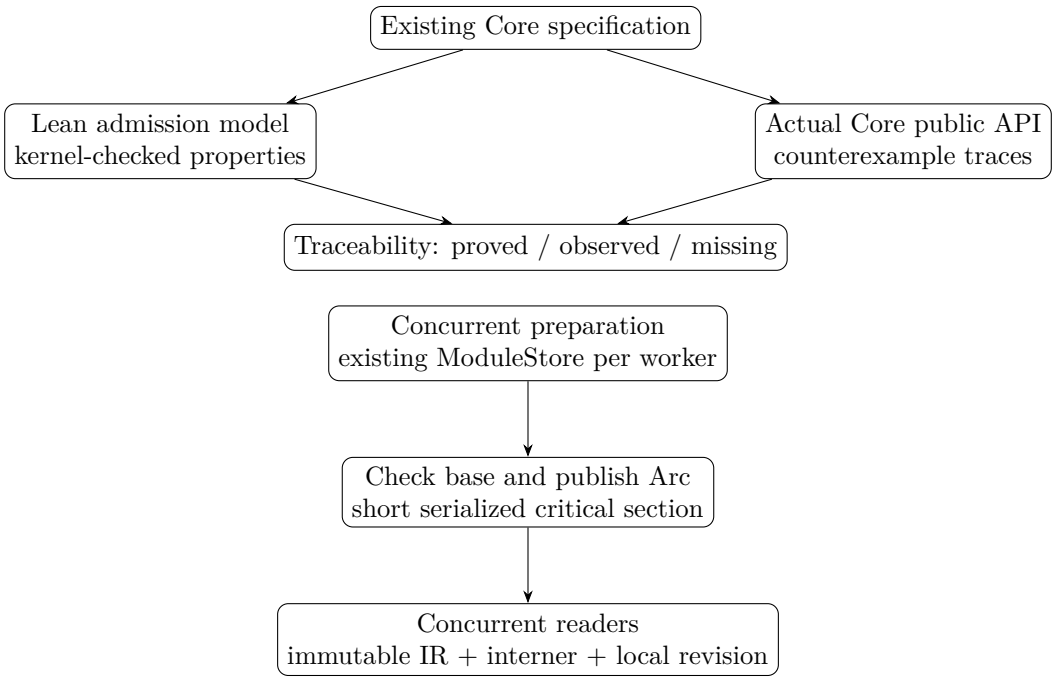
Created-at: 2026-09-13 17:05 +02:00. ETA: 20–35 minutes for the additive API, bounded concurrency regressions, proof model, gates and archival. The user authorized continuing the parallel execution model. This scopes the protected Core edit to the module-store execution boundary and its tests; no dependency change, compiler rewrite, benchmark or profiling is authorized.

Reuse the current checkout at 67741f8 (fresh origin and GitHub main match). Preserve other workers' benchmark source/manifest and plan edits. Core owns compiled model snapshots and publication; consumers still own worker scheduling. The existing ModuleStore remains the only compilation implementation. A fresh store per preparation avoids its root-only cache returning old source. Move its interner together with the compiled program, retaining consistent symbol identity.

New source: `hymeko_core/src/module_store/execution.rs` and `execution_tests.rs`; export through the existing module store. API: CompiledModel, ExecutionSnapshot, ExecutionState, PreparedCompilation, PrepareError, PublicationError. Preparation compiles outside the publication lock. Commit consumes an opaque candidate, rejects foreign state and stale base, checks revision overflow, then replaces one Arc under a short write lock. Readers clone an Arc under a read lock and run outside it. Retained snapshots keep the IR and interner alive. No implicit retry, merging, latest-request-wins policy or fairness guarantee. A revision is local publication order, not a distributed version vector.

Verification: only the new library test namespace; same-base concurrent writers, reader lifetime and reentrant publication, independent states, failed preparation, fresh source at the same path, poison and overflow. Compare compiled semantics with the existing ModuleStore and full semantic digest (not structural V1). Extend the existing Lean model with local scalar-revision admission laws; it is an abstraction, not a Rust memory-model or compiler-correctness proof. Run targeted rustfmt/Clippy and portable symbol/duplication/preflight gates. Preserve baseline and report external gate interference explicitly.

Other affected files: these four plan artifacts, the existing `docs/verification/unified-ir/distributed-semantics.lean`, generated `SYMBOLS.md`, and ignored `reports/2026-09-13-parallel-executio`
Rollback only this task's named source and documentation edits. No commit/push requested. Memory risks: concurrent compilation and retained readers multiply live models; there is no automatic queue/backpressure, use bounded consumer workers. Tests use small fixed inputs under the host's 16 GiB ceiling. No performance measurement.


Existing Core specification
Lean admission model
kernel-checked properties
Actual Core public API
counterexample traces
Traceability: proved / observed / missing
Concurrent preparation
existing ModuleStore per worker
Check base and publish Arc
short serialized critical section
Concurrent readers
immutable IR + interner + local revision